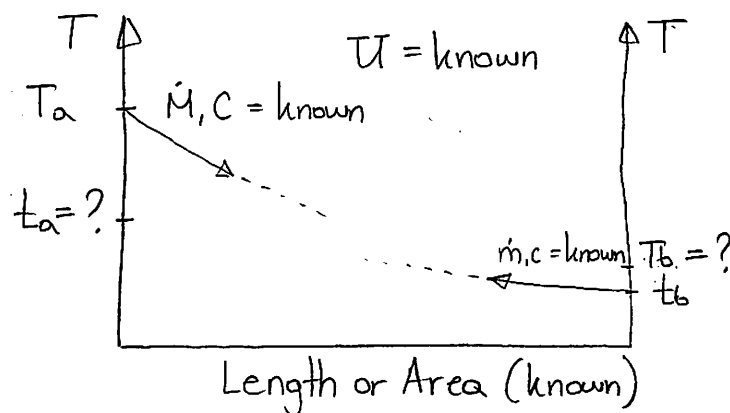


In general, we can say the following:

Parallel Flow	Counter Flow
<u>Disadvantage #1</u>: Large temp. difference at one end of the heat exchanger causes added thermal stresses & early failure <u>Disadvantage #2</u>: The outlet temp. of the cold fluid never exceeds the outlet temp of the hot fluid. Less efficient.	<u>Advantage #1</u>: More uniform ΔT minimizes the thermal stresses throughout the exchanger. <u>Advantage #2</u>: The outlet temp of the cold fluid can approach the highest temperature of the hot fluid. More efficient. <u>Advantage #3</u>: More uniform ΔT produces a more uniform q.

ϵ -NTU Method (Effectiveness - NTU method)

Note, in most heat exchanger design problems, we don't know the fluid outlet temperatures, i.e. $T_{1,out}$ or $T_{2,out}$



Now we can define a quantity called the capacity

$$C_h = (\dot{m}C)_{\text{hot}} \text{ [W/K]}; \quad C_c = (\dot{m}C)_{\text{cold}} \text{ [W/K]}$$

To solve, we could guess an exit temp., solve for $Q_h = Q_c = C_h \Delta T_h = C_c \Delta T_c$

Then we would have to calculate Q from $UA LMTD$ and check against our previous answer. If differing, we would guess another exit temp. and try again.

We can make our lives much easier with ϵ -NTU.

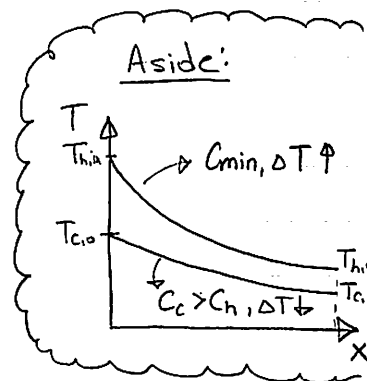
$$\epsilon = \frac{\text{actual heat transferred}}{\left(\begin{array}{l} \text{maximum heat that could possibly} \\ \text{be transferred from one stream to the other} \end{array} \right)}$$

Mathematically, this is equal to:

$$\epsilon = \frac{C_h (T_{h,in} - T_{h,out})}{C_{min} (T_{h,in} - T_{c,in})} = \frac{C_c (T_{c,out} - T_{c,in})}{C_{min} (T_{h,in} - T_{c,in})}$$

where C_{min} is the smaller of C_h & C_c .

The reason we use C_{min} is because the fluid with the lower capacity will be the one which undergoes the maximal temperature change, i.e. $q = C_{min} \Delta T_{max}$, $C_{min} \uparrow, \Delta T \downarrow$



So we can write: $Q = \epsilon C_{min} (T_{h,in} - T_{c,in})$

We can also define: $NTU = \frac{UA}{C_{min}} = \frac{\text{heat rate capacity of h.x.}}{\text{heat capacity rate of flow}}$
(dimensionless)

Using energy balances & simplifying, we can solve for our two cases:

Parallel Flow:

$$-\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h}\right) NTU = \ln \left[-\left(1 + \frac{C_c}{C_h}\right) \epsilon \frac{C_{min}}{C_c} + 1 \right]$$

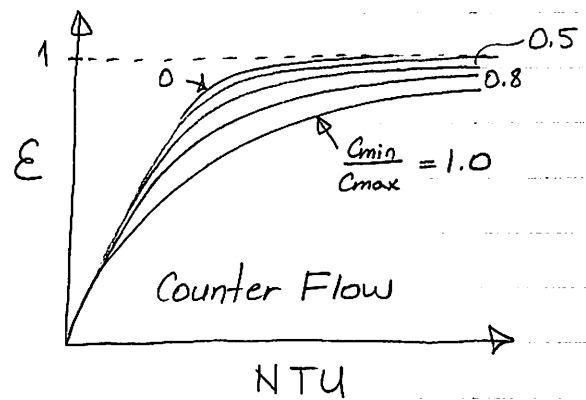
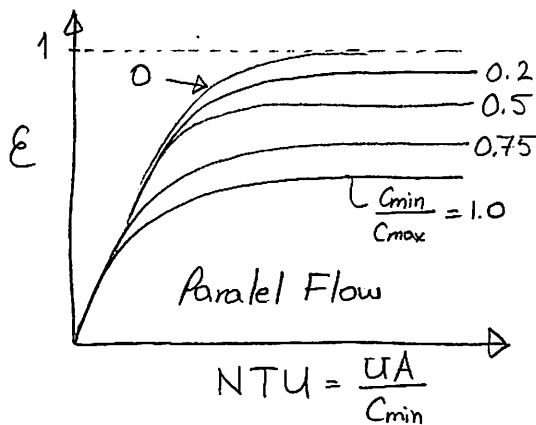
Solving for ϵ , we obtain:

$$\epsilon = \frac{1 - \exp \left[-\left(1 + C_{min}/C_{max}\right) NTU \right]}{1 + C_{min}/C_{max}} = f \left(\frac{C_{min}}{C_{max}}, NTU \text{ only} \right)$$

Counter Flow:

$$\epsilon = \frac{1 - \exp[-(1 + C_{\min}/C_{\max})NTU]}{1 - (C_{\min}/C_{\max})\exp[-(1 - C_{\min}/C_{\max})NTU]} = f\left(\frac{C_{\min}}{C_{\max}}, NTU, \text{only}\right)$$

We can plot our results in a more useful form



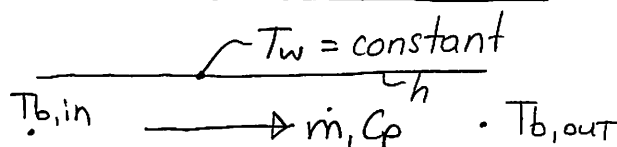
Note, the ϵ -NTU method can be applied to any arbitrarily shaped heat exchanger with more difficult flow patterns (i.e. cross flow, mixed flow, etc...)

For a great book, see "Compact Heat Exchangers", Kays & London, (1955.)

Note that our solution for the pipe flow with constant wall temperature will also simplify to this:

$$\underbrace{\frac{T_{b,out} - T_{b,in}}{T_w - T_{b,in}}}_{\epsilon} = 1 - e^{\underbrace{\left(-\frac{hPL}{\dot{m}C_p}\right)}_{\substack{hPL = UA \\ \dot{m}C_p = C_{\min}}}} \Rightarrow \text{See page 120 of notes. } P = \text{perimeter}$$

$$\boxed{\epsilon = 1 - e^{-NTU}} \Rightarrow f(NTU \text{ only})$$



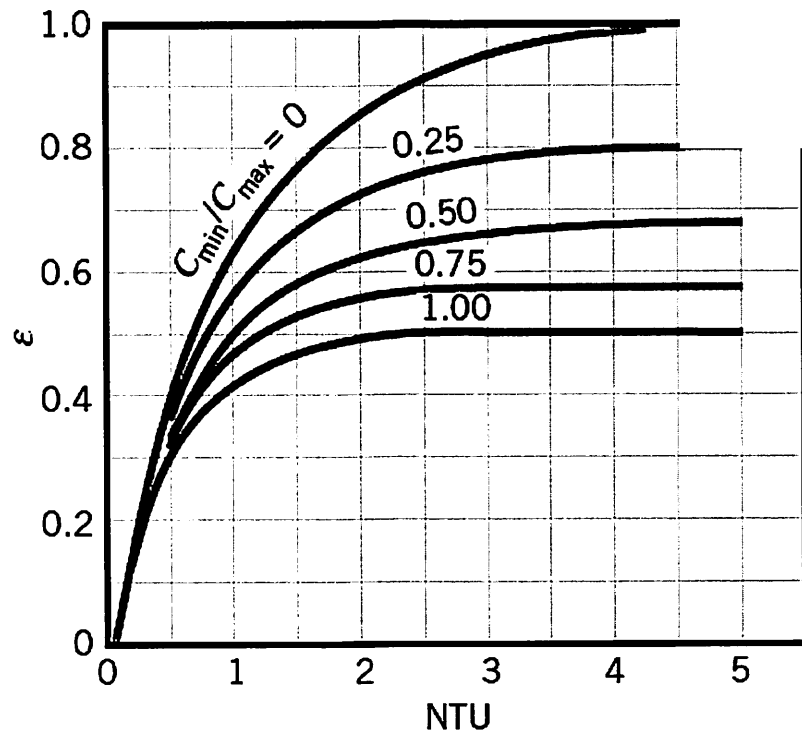


FIGURE 11.10 Effectiveness of a parallel-flow heat exchanger (Equation 11.28).

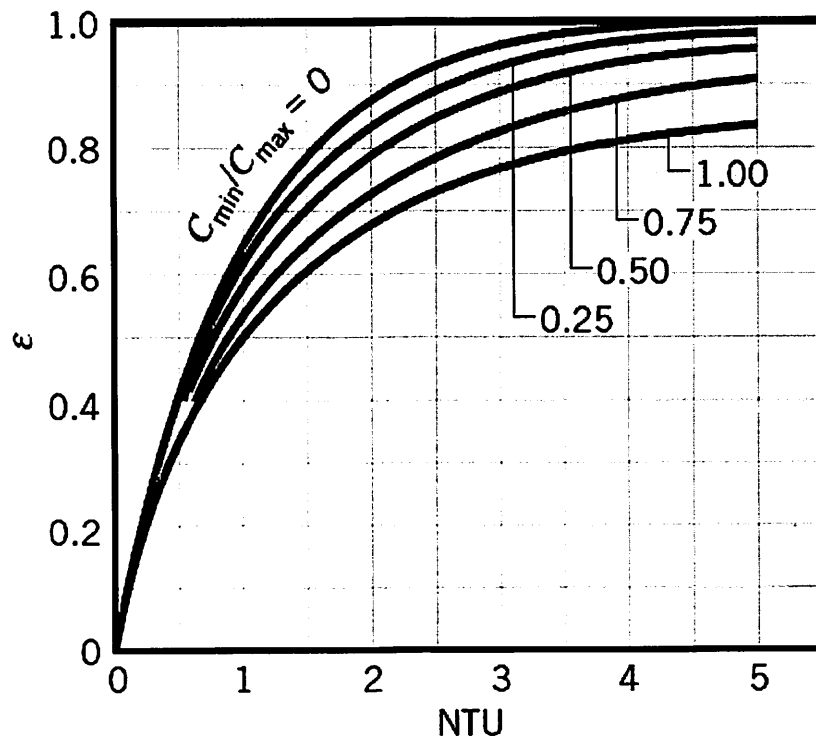


FIGURE 11.11 Effectiveness of a counterflow heat exchanger (Equation 11.29).

Example | Consider the following parallel-flow h.x. specification

Cold flow enters at 40°C , $C_c = 20,000 \text{ W/K}$

Hot flow enters at 150°C , $C_h = 10,000 \text{ W/K}$

$$A = 30 \text{ m}^2, U = 500 \text{ W/m}^2 \cdot \text{K}$$

Determine the heat transfer and the exit temperatures.

Here, we can't use LMTD since we don't have exit conditions so ϵ -NTU must be used.

$$NTU = \frac{UA}{C_{\min}} = \frac{500(30)}{10,000} = 1.5$$

$$\frac{C_{\min}}{C_{\max}} = 0.5$$

From our tables, we obtain $\epsilon = 0.596$

$$Q = \epsilon C_{\min} (T_{h,\text{in}} - T_{c,\text{in}}) = 0.596(10,000)(110) \\ = 655,600 \text{ W} = 655.6 \text{ kW}$$

From energy balances:

$$T_{h,\text{out}} = T_{h,\text{in}} - \frac{Q}{C_h} = 150 - \frac{655,600}{10,000} = 84.44^\circ\text{C}$$

$$T_{c,\text{out}} = T_{c,\text{in}} + \frac{Q}{C_c} = 40 + \frac{655,600}{20,000} = 72.78^\circ\text{C}$$

Example | Suppose we had the same problem, but A = unknown and we want $T_{h,\text{out}} = 90^\circ\text{C}$. Calculate A .

Here we can use the ϵ -NTU method or LMTD since we have an exit temperature. Applying an energy balance:

$$T_{c,\text{out}} = T_{c,\text{in}} + \frac{C_h}{C_c} (T_{h,\text{in}} - T_{h,\text{out}}) = 40 + \frac{1}{2}(150 - 90) = 70^\circ\text{C}$$

Using the ϵ -NTU method:

$$\epsilon = \frac{C_h (T_{h,in} - T_{h,out})}{C_{min} (T_{h,in} - T_{c,in})} = \frac{10,000 (150 - 90)}{10,000 (150 - 40)} = 0.5455$$

From our charts, $NTU \cong 1.15 = UA/C_{min}$

$$A = \frac{10,000 (1.15)}{500} = 23.00 m^2$$

We could have also used LMTD (note doesn't matter which side is a and b)

$$\begin{aligned} LMTD &= \frac{\Delta T_a - \Delta T_b}{\ln \left(\frac{\Delta T_a}{\Delta T_b} \right)} = \frac{(T_{h,in} - T_{c,in}) - (T_{h,out} - T_{c,out})}{\ln \left(\frac{T_{h,in} - T_{c,in}}{T_{h,out} - T_{c,out}} \right)} \\ &= \frac{(150 - 40) - (90 - 70)}{\ln \left(\frac{150 - 40}{90 - 70} \right)} = 52.79 K \end{aligned}$$

So from $Q = UA(LMTD)$

$$A = \frac{Q}{U(LMTD)} = \frac{10,000 (150 - 90)}{(500)(52.79)} = 22.73 m^2$$

The answers differ by 1%, which reflects graph reading innacuracy.

Example / On page 109 of the notes, we did external flow with variable wall temperature. We solved for:

$$q''|_{y=0} = 0.332 \left(\frac{k}{x}\right) Re_x^{1/2} Pr^{1/3} \int_0^x \frac{d(T_w - T_\infty)}{d\zeta} \frac{d\zeta}{\left[1 - \left(\frac{\zeta}{x}\right)^{3/4}\right]^{1/3}}$$

Solve for $q''|_{y=0}$ if $(T_w - T_\infty) = \beta \sqrt{x}$

Since it's a continuous temperature change, $\zeta \approx x$

$$(T_w - T_\infty) = \beta \sqrt{\zeta}$$

$$\frac{d(T_w - T_\infty)}{d\zeta} = \frac{1}{2} \beta \frac{1}{\sqrt{\zeta}}$$

Let $s = \frac{\zeta}{x} \Rightarrow q''|_{y=0} = 0.332 \left(\frac{k}{x}\right) Re_x^{1/2} Pr^{1/3} \beta \cdot \frac{1}{2} \int_0^x \frac{d\zeta}{\zeta^{1/2} \left[1 - \left(\frac{\zeta}{x}\right)^{3/4}\right]^{1/3}}$

Multiply by $\left(\frac{x}{x}\right)$, we will obtain

$$q''|_{y=0} = 0.332 \left(\frac{k}{x}\right) Re_x^{1/2} Pr^{1/3} x^{1/2} \beta \cdot \frac{1}{2} \int_0^1 \frac{ds}{s^{1/2} \left[1 - s^{3/4}\right]^{1/3}}$$

$$\int_0^1 \frac{ds}{s^{1/2} \left[1 - s^{3/4}\right]^{1/3}}, \quad \lambda = (1 - s^{3/4})$$

$$s = (1 - \lambda)^{4/3}$$

$$s^{1/2} = (1 - \lambda)^{2/3}$$

$$\left[1 - s^{3/4}\right]^{1/3} = \lambda^{1/3}$$

We need to solve this

$$= \frac{4}{3} \int_0^1 \frac{(1 - \lambda)^{1/3} d\lambda}{(1 - \lambda)^{2/3} \lambda^{1/3}} = \frac{4}{3} \int_0^1 \frac{d\lambda}{\lambda^{1/3} (1 - \lambda)^{1/3}} = \frac{4}{3} \int_0^1 x^{\frac{2}{3}-1} (1 - \lambda)^{\frac{2}{3}-1} d\lambda$$

So we know from integral tables: $\int_0^1 x^{p-1} (1-x)^{q-1} dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$

$$q''|_{y=0} = 0.332 \left(\frac{k}{x}\right) Re_x^{1/2} Pr^{1/3} x^{1/2} \beta \underbrace{\frac{1}{2} \cdot \frac{4}{3}}_{\Delta T(x)} (2.06) = \frac{(1.355)^2}{0.893} = 2.06$$

$$Nu_x = \left(\frac{q''|_{y=0} \cdot x}{\Delta T(x) \cdot k_f} \right) = 0.455 Re_x^{1/2} Pr^{1/3}$$

$\Rightarrow$ Note, we get constant heat flux only if $T(x) = \beta \sqrt{x}$